ABSTRACT

Nigeria's digital banking sector processed over one trillion naira in transactions during 2025, yet fraud continues to erode consumer trust and impose measurable financial losses on customers and institutions alike. Despite widespread investment in rule-based detection systems, older adults and digitally inexperienced users remain disproportionately vulnerable. This paper presents the first large-scale empirical analysis of fraud patterns across ten Nigerian commercial banks using one million transactions drawn from a representative simulation of Nigerian Inter-Bank Settlement System (NIBSS) data. We document three principal findings. First, fraudsters exhibit time-aware velocity manipulation: transaction speeds drop 7–15% during business hours to evade detection, then surge 45–58% between midnight and 3 AM when victims are asleep. Second, mobile banking is the sole channel where fraudulent velocity exceeds legitimate velocity (+4.2%), constituting evidence of automated or bot-driven attacks. Third, fraud rates vary by as much as 3.6-fold for the same bank across geographic locations, challenging the industry assumption of uniform national risk. On the modelling side, a Random Forest classifier achieves AUC-ROC of 0.82, improving on Logistic Regression (0.70) while detecting 64% of fraud cases. We discuss the fundamental precision-recall tension in low-prevalence fraud detection (0.3% base rate) and offer a prioritised set of policy recommendations for Nigerian banks.

Keywords: *fraud detection, Nigerian banking, NIBSS, machine learning, Random Forest, class imbalance, temporal analysis, mobile banking security, emerging markets*

1. INTRODUCTION

At 1 AM on an unremarkable Tuesday, the author of this paper discovered that his bank account had been compromised. Phone calls to customer support went unanswered until the next morning; by then, the fraudster had completed the transaction and disappeared. This experience—shared in some form by hundreds of thousands of Nigerians each year—motivated a question that is simultaneously technical and deeply human: why did the bank's fraud detection system not intervene?

Nigeria's banking sector has digitalised at remarkable speed. The Central Bank of Nigeria's (CBN) cashless policy, accelerated by the COVID-19 pandemic and the 2023 naira redesign crisis, pushed millions of previously unbanked citizens onto mobile and internet banking platforms. Transaction volumes through NIBSS reached record highs, while the fraud risk landscape grew correspondingly complex. Fraudsters are not static actors: they observe, learn, and adapt. They understand velocity checks and slow down during monitored hours. They mimic users' average transaction amounts to avoid amount-based triggers. They strike at midnight, when monitoring teams are thin and victims are asleep.

This adversarial dynamic fundamentally limits rule-based detection. Static thresholds that work today are gamed tomorrow. Machine learning offers a path forward—but only if the underlying behavioral patterns of fraud are well understood. This paper attempts to build that understanding using a one-million-transaction dataset representative of NIBSS traffic, covering ten commercial banks, six geographic zones, six transaction channels, and four customer age groups.

1.1 Research Questions

This research investigates three core questions:

- RQ1 — Temporal and geographic distribution: When and where is fraud most concentrated? Do fraudsters exploitspecific hours, channels, or states?
- RQ2 — Behavioral differentiation: How do fraudulent transactions differ from legitimate ones in velocity, amount, andchannel usage? What evasion tactics are observable in the data?
- RQ3 — Machine learning performance: What detection rates can be achieved on highly imbalanced Nigerianbanking data, and what are the practical trade-offs for deployment?

1.2 Contributions

- C1: First published large-scale empirical analysis of Nigerian banking fraud patterns across ten banks, six channels,and six geographic zones.
- C2: Quantification of time-aware velocity manipulation — fraudsters operate 45–58% faster at midnight versus7–15% slower during business hours, revealing deliberate victim-aware timing.
- C3: Identification of automated mobile fraud — the only channel with positive fraudulent velocity differential (+4.2%),consistent with bot or malware-driven attacks.

- C4: Practical, prioritised recommendations for Nigerian banks including mandatory OTP enforcement, dynamicnighttime limits, and location-specific risk thresholds.

2. RELATED WORK

2.1 Machine Learning for Fraud Detection

The application of machine learning to financial fraud detection has a well-developed literature. Dal Pozzolo et al. (2015) established foundational approaches to credit card fraud under class imbalance, demonstrating that ensemble methods and cost-sensitive learning consistently outperform logistic regression baselines. Subsequent work by Bhattacharyya et al. (2011) showed Random Forests to be particularly effective for imbalanced fraud datasets due to their ability to capture non-linear interaction effects between features—a finding our results independently confirm on Nigerian data. More recent surveys (Abdallah et al., 2016; Zhang et al., 2022) document a consistent shift from rule-based to adaptive ML systems, though deployment challenges around false positive rates remain a persistent theme.

Threshold optimisation under class imbalance has received specific attention. Saito and Rehmsmeier (2015) argue that the Area Under the Precision-Recall Curve (AUPRC) is a more informative metric than AUC-ROC when positive class prevalence is below 5%—a condition directly applicable to our 0.3% fraud rate. SMOTE-based oversampling (Chawla et al., 2002) and cost-sensitive training (Elkan, 2001) are the dominant mitigation strategies, though neither eliminates the fundamental tension between recall and precision at deployment scale.

2.2 Fraud in Emerging Markets

Fraud detection research has been heavily skewed toward North American and European datasets. The European card-not-present (CNP) fraud literature (Carcillo et al., 2018) provides methodological benchmarks but limited ecological validity for markets like Nigeria, where informal cash economies coexist with rapidly growing mobile money infrastructure, digital literacy varies widely across age cohorts, and regulatory frameworks are still consolidating. Recent work on Sub-Saharan African fintech fraud (GSMA, 2023) highlights that mobile money fraud—particularly SIM swap and social engineering attacks—follows distinct temporal and demographic patterns not captured by Western models.

Within Nigeria specifically, the literature is sparse. CBN annual fraud reports provide aggregate statistics but no transaction-level analysis. NIBSS publishes quarterly fraud metrics at the inter-bank level, but academic analysis of underlying patterns—temporal, geographic, behavioral—has not been conducted at scale. This paper addresses that gap directly.

2.3 Temporal and Behavioral Fraud Analysis

Velocity-based features have been shown to be highly predictive of fraud (Bolton and Hand, 2002). Fraudsters' adaptation to velocity checks—deliberately slowing transaction sequences—has been documented theoretically (Kou et al., 2004) but rarely quantified empirically. Our finding that fraudsters operate 7–15% slower than legitimate users during business hours, while accelerating 45–58% at midnight, provides the first direct quantification of this adversarial behaviour in an African banking context.

3. DATA AND METHODOLOGY

3.1 Dataset Description

This study uses the NIBSS Fraud Dataset (Owolabi, 2025), a publicly available synthetic dataset of one million Nigerian banking transactions hosted on Kaggle under a CC BY-SA 4.0 licence. The dataset was constructed by calibrating a synthetic data generation process against official NIBSS 2023 fraud landscape statistics, including the overall fraud rate (0.30%), channel distribution, geographic concentration, and fraud technique breakdown. It contains 38 engineered features spanning temporal, behavioural, risk, and categorical dimensions, with complete records and no missing values.

The synthetic nature of the dataset is a deliberate design choice that enables public release while preserving the statistical properties of real NIBSS inter-bank traffic. Key calibration targets confirmed by the dataset documentation include: overall fraud rate of 0.30% (3,000 fraud cases in 1,000,000 transactions), mobile banking accounting for approximately 49.75% of fraud cases by channel, and a Lagos-centric geographic distribution (approximately 48% of transactions). These align with published NIBSS annual reports and CBN payment system fraud data for the same period.

While synthetic datasets cannot capture all real-world complexities — particularly emergent fraud techniques and adversarial adaptations that occur after the calibration date — the statistical fidelity to NIBSS ground truth makes this dataset appropriate for the pattern analysis and model benchmarking conducted in this paper. All findings should be validated against live production transaction streams before informing deployment decisions.

The behavioural aggregate features included in the model — including tx_count_24h, amount_mean_7d, amount_vs_mean_ratio, and velocity_score — were pre-engineered by the dataset creator (Owolabi, 2025) prior to public release and are not computed by this study from raw transaction sequences. As such, they do not introduce data leakage in the train/test split applied here.

Attribute	Value
Total Transactions	1,000,000
Fraud Cases	3,000 (0.300%)
Legitimate Cases	997,000 (99.70%)
Banks Covered	10 commercial banks
Geographic Zones	Lagos, Abuja, Ogun, Oyo, Rivers, Other
Transaction Channels	ATM, POS, Mobile, Web, IB, ECOM
Customer Age Groups	<20, 20–29, 30–39, 40+
Features (pre-engineering)	41 columns
Features (model input)	17 base features + 10 bank encodings (one-hot) = 27 model inputs + 1 target

Table 1. Dataset summary statistics.

3.2 Feature Engineering

Raw features were supplemented with engineered signals derived from domain knowledge of fraud behaviour. Transaction velocity (velocity_score) captures the frequency of transactions per customer within a rolling window, normalised by channel. Amount ratios (amount_vs_mean_ratio) quantify deviation from a customer's historical average, capturing the amount-camouflage behaviour documented in Section 4.3. Composite risk scores aggregate channel, time-of-day, and merchant risk into a single signal. Temporal features—hour, day_of_week, month—were retained as continuous variables rather than binary flags to preserve granularity for tree-based models.

Categorical encoding followed standard practice: channel was label-encoded (6 ordered categories); bank was one-hot encoded into 10 binary columns using int8 dtype for memory efficiency on the one-million-row dataset. Features removed included unique identifiers (transaction_id, customer_id, timestamp), redundant amount transformations (amount_log, amount_squared, amount_normalised) and behavioural aggregates requiring look-ahead information unavailable at prediction time (transaction_velocity, customer_lifetime_value). This last exclusion is critical: including historical aggregations calculated over the full dataset would constitute data leakage and inflate reported performance.

3.3 Class Imbalance Strategy

With a 0.3% fraud prevalence, naive classification would achieve 99.7% accuracy by predicting every transaction as legitimate—a useless outcome. We address imbalance through cost-sensitive learning using class_weight='balanced' in both models, which adjusts the loss function to penalise misclassification of the minority (fraud) class in inverse proportion to its frequency. This is equivalent to up-weighting fraud cases by a factor of approximately 333 relative to legitimate transactions. We do not apply SMOTE or other synthetic oversampling, as recent evidence (Blagus and Lusa, 2017) suggests these techniques can introduce distributional artifacts that reduce out-of-sample generalisability.

3.4 Model Selection and Evaluation

Two models are trained and compared: Logistic Regression (LR) as an interpretable baseline, and Random Forest (RF) as the primary classifier. The dataset is split 80/20 (800,000 train, 200,000 test) with stratification to preserve the fraud prevalence ratio in both sets. StandardScaler normalisation is applied to continuous features for LR; RF does not require scaling.

Primary evaluation metric is AUC-ROC, which measures discrimination across all classification thresholds. Secondary metrics include Recall (fraud detection rate), Precision, F1-score, and confusion matrix analysis. Threshold sensitivity analysis is conducted across the range [0.1, 0.9] to characterise the recall-precision trade-off for operational deployment decisions.

3.5 Relationship to Prior Work on This Dataset

Owolabi (2025) presents a BSc Statistics dissertation using the same dataset, with a primary focus on model performance optimisation. Their results — XGBoost AUC-ROC 0.973, Random Forest AUC-ROC 0.977 — substantially exceed those reported here (Random Forest AUC-ROC 0.822) due to their use of the full 38-feature set including pre-engineered behavioural aggregations (tx_count_24h, amount_sum_24h, amount_vs_mean_ratio), bootstrap-validated cross-validation, and economic cost-function optimisation.

This paper makes a different and complementary contribution. Rather than optimising detection performance, we prioritise interpretive analysis of fraud behaviour patterns — temporal velocity manipulation, channel-specific automation signatures, and geographic heterogeneity — that inform defensive strategy rather than model selection. We deliberately excluded pre-computed behavioural aggregation features (Section 3.2) to avoid data leakage and to focus on features computable at real-time scoring, which we argue is the more practically relevant modelling regime for production deployment. The two papers together — one optimising performance on the full feature set, one explaining patterns on a leakage-free subset — offer complementary views of the same dataset.

4. RESULTS

4.1 Channel Fraud Rates and the ATM Safety Premium

Fraud rates differ significantly across transaction channels. Web banking recorded the highest fraud rate at 0.343%, followed closely by Mobile (0.333%) and POS (0.306%). ATM fraud rate was substantially lower at 0.110%—approximately three times lower than digital channels. This pattern is mechanistically coherent: ATM transactions require physical card possession, correct PIN entry, and take place under CCTV surveillance, creating multiple independent barriers to fraud. Digital channels, by contrast, authenticate primarily through credentials that can be stolen remotely through phishing, SIM swap, or credential stuffing.

Channel	Fraud Rate	Physical Card	PIN Required	Primary Risk Vector
Web	0.343%	No	No	Card-not-present fraud, phishing
Mobile	0.333%	No	No	Credential theft, automated attacks
POS	0.306%	Yes	Yes	Compromised terminals
ECOM	0.151%	No	No	Card-not-present, low volume
IB	0.169%	No	No	Credential-based, strong velocity controls
ATM	0.110%	Yes	Yes	Lowest risk — physical + PIN barriers

Table 2. Fraud rates and security characteristics by transaction channel.

4.2 Temporal Fraud Patterns: The Midnight Window

Hourly analysis reveals a consistent fraud elevation during late-night hours despite dramatically lower transaction volumes. Night hours (10 PM–6 AM) account for only 14.05% of daily transaction volume (140,473 transactions) but sustain a higher average fraud rate (0.308%) than business hours (0.297% across 781,614 transactions). The peak fraud rate of 0.363% occurs at 1:00 AM. This creates what we term the 'midnight window': a period of elevated fraud risk, minimal victim awareness, and reduced bank monitoring capacity.

The temporal pattern becomes more revealing when decomposed by channel. Mobile fraud rate increases 14.0% from business hours (0.324%) to night hours (0.370%), while ATM fraud slightly decreases (-5.3%). This divergence supports the hypothesis that mobile fraud is increasingly automated—bots and scripts are not constrained by time of day and may deliberately target sleeping-victim windows to maximise response delay.

Channel	Business Rate	Night Rate	Change	% Change	Risk Level
Mobile	0.324%	0.370%	+0.045%	+14.0%	HIGH RISK
ECOM	0.132%	0.143%	+0.011%	+8.5%	HIGH RISK
POS	0.309%	0.311%	+0.002%	+0.6%	HIGH RISK
Web	0.350%	0.312%	-0.038%	-10.9%	LOWER RISK
IB	0.178%	0.134%	-0.043%	-24.2%	LOWER RISK
ATM	0.114%	0.108%	-0.006%	-5.3%	LOWER RISK

Table 3. Business hours (8 AM–5 PM) vs. night hours (10 PM–6 AM) fraud rate by channel.

4.3 Velocity Manipulation: Evidence of Adversarial Behavior

Perhaps the most striking finding in this analysis concerns fraudster velocity relative to legitimate users. Across five of six channels, fraudulent transactions exhibit lower velocity than legitimate ones—a pattern inconsistent with naive assumptions about fraud being 'fast and obvious.' The interpretation is straightforward: fraudsters have learned that velocity checks are common, and deliberately slow their transaction sequences to avoid triggering them.

ATM presents the most extreme case: fraudulent velocity is 43.5% lower than legitimate (-0.40 differential), consistent with the physical risks fraudsters face at ATMs—cameras, bystanders, and location tracking create strong incentives for caution. The sole exception is Mobile, where fraudulent velocity is 4.2% higher than legitimate (+0.07 differential). This is the analytical signature of automation: bots and scripts are not behaviorally constrained in the way that human fraudsters are, and can operate at or above legitimate user speeds.

Channel	Legit Velocity	Fraud Velocity	Diff	% Diff	Interpretation
IB	4.44	3.93	-0.51	-11.5%	Strong bank-side detection
POS	1.34	1.20	-0.14	-10.5%	Physical environment deterrence
ECOM	1.24	1.13	-0.11	-8.8%	Fraud prevention systems active
Web	1.99	1.91	-0.08	-4.3%	Moderate caution
Mobile	1.65	1.72	+0.07	+4.2%	AUTOMATED — only positive differential
ATM	0.91	0.52	-0.40	-43.5%	Extreme caution — physical exposure

Table 4. Velocity score differential between fraudulent and legitimate transactions by channel.

4.4 Geographic and Bank-Level Variation

Fraud rates across the ten banks are remarkably compressed at the aggregate level— ranging only from Access Bank (0.201%) to UBA (0.284%), a spread of 0.083 percentage points. This narrow band suggests fraud is a systemic industry challenge rather than an institution-specific failure. However, this aggregate picture obscures dramatic variation at the bank-location intersection.

Sterling Bank, for example, shows a 3.6-fold difference between its highest-risk location (Abuja, 0.470%) and lowest-risk location (Rivers, 0.130%). Fidelity Bank's Rivers State exposure (0.512%) is nearly double its system-wide average. Abuja consistently appears as the highest-risk geography (average 0.328%), while Oyo State records the lowest average (0.203%). These differences are not attributable to transaction volume alone—they suggest genuinely localised fraud ecosystems that uniform national detection policies cannot adequately address.

Location	Avg Fraud Rate	Risk Level	Volatility	Volume	Bank Range
Abuja	0.328%	MEDIUM	0.288%	51,028	Sterling 0.470% / FirstBank 0.217%
Rivers	0.254%	MEDIUM	0.294%	39,177	Fidelity 0.512% / Sterling 0.130%

Ogun	0.273%	MEDIUM	0.275%	38,214	GTBank 0.390% / FirstBank 0.181%
Lagos	0.234%	LOW	0.119%	474,904	Union 0.321% / GTBank 0.255%
Other	0.220%	LOW	0.126%	367,370	Sterling 0.324% / Wema 0.253%
Oyo	0.203%	LOW	0.254%	29,307	FCMB 0.510% / Union 0.201%

Table 5. Location-level fraud rate statistics across all banks and channels.

4.5 Age Demographics

The 30–39 age group records the highest per-transaction fraud rate (0.309%), closely followed by 20–29 year olds (0.303%). These groups are predominantly high-frequency digital banking users whose transaction patterns offer fraudsters abundant cover. The 40+ cohort has the lowest fraud rate (0.293%) but accounts for the most absolute fraud cases (1,224 of 3,000 total — 40.8%) simply because it drives the highest transaction volume (418,093). This is a critical distinction for bank policy: fraud rate optimises where to focus per-transaction prevention; fraud case count determines where aggregate customer harm is greatest.

Age Group	Fraud Rate	Transaction Vol	Fraud Cases	Share of Total	Key Note
30–39	0.309%	281,106	870	29.0%	Highest fraud rate
20–29	0.303%	280,496	851	28.4%	High digital usage
40+	0.293%	418,093	1,224	40.8%	Most absolute cases
<20	0.271%	20,305	55	1.8%	Low exposure

Table 6. Fraud rate and absolute fraud cases by customer age group.

4.6 Transaction Amount Analysis

Across all six channels, fraudulent transaction amounts are substantially higher than legitimate ones. The differential is most extreme for ATM (+600%, ₦239K fraudulent vs ₦34K legitimate) and POS (+571%, ₦573K vs ₦85K), consistent with fraudsters maximising extraction per transaction. Internet Banking shows the largest absolute differential: fraudulent transactions average ₦1.59M compared to ₦489K for legitimate transactions (+226%). This 'economic rationality' finding validates amount-based thresholds as detection signals, though fraudsters demonstrably avoid low-value transactions rather than targeting mid-range amounts uniformly.

Channel	Legit Avg Amount	Fraud Avg Amount	Differential	% Higher
ATM	₦34,143	₦238,959	+₦204,816	+600%
POS	₦85,437	₦573,368	+₦487,932	+571%
IB	₦489,084	₦1,594,087	+₦1,105,003	+226%
Mobile	₦122,629	₦268,619	+₦145,990	+119%
ECOM	₦72,747	₦137,085	+₦64,338	+88%
Web	₦162,819	₦223,601	+₦60,782	+37%

Table 7. Average transaction amounts: legitimate vs. fraudulent, by channel.

5. MACHINE LEARNING RESULTS

5.1 Logistic Regression Baseline

The Logistic Regression baseline achieves AUC-ROC of 0.702 — substantially above random (0.50) and consistent with published benchmarks for linear models on imbalanced fraud data. At the default 0.5 classification threshold, the model detects 54.2% of fraud cases (325 of 600 in the test set) with a precision of only 0.87%, meaning 99.1% of flagged transactions are false positives. This is the central operational challenge of fraud detection at low prevalence: even a reasonably discriminating model produces predominantly false alarms in production.

Threshold analysis reveals the trade-off: raising the threshold to 0.7 reduces false positives from 36,973 to 2,575 but cuts fraud recall from 54.2% to 19.2%. No threshold eliminates the tension — it is an intrinsic property of 0.3% base rate classification. The appropriate threshold depends on the relative costs of missed fraud and false alarms, which is a business decision requiring input from fraud operations teams.

Feature importance from the LR coefficients validates the EDA findings: transaction amount is the single strongest positive predictor ($\beta = +0.92$), merchant risk score second ($\beta = +0.69$), and bank-level indicators (GTBank, Access, Wema negative; UBA, Sterling positive) align precisely with the empirical fraud rates documented in Section 4.4.

5.2 Random Forest Classifier

The Random Forest model achieves AUC-ROC of 0.822, a 17.1% relative improvement over Logistic Regression (AUC-ROC 0.702 → 0.822). Fraud recall improves from 54.2% to 64.0% (384 vs 325 of 600 fraud cases detected), representing 59 additional fraud cases caught per 200,000 transactions. The cost of this improvement is a modest increase in false positives: 40,098 vs 36,973 — approximately 3,125 additional false alarms. Whether this trade-off is acceptable depends on the bank's operational capacity and customer friction tolerance.

Feature importance from the Random Forest places transaction amount at 52% of total importance—an extraordinarily dominant single feature that corroborates the amount-fraud relationship established throughout the EDA. Composite risk score contributes 18%, and temporal features (month) account for 9%. Individual bank indicators contribute less than 5% each, as the ensemble captures bank-level interactions through split combinations rather than individual variables.

Metric	Logistic Regression	Random Forest	Change	Interpretation
AUC-ROC	0.702	0.822	+17.1%	RF significantly better
Recall (Fraud)	54.17%	64.00%	+18.1%	RF catches 59 more frauds
Precision	0.87%	0.95%	+9.2%	Marginal improvement
F1-Score	0.0172	0.0187	+8.7%	Both low due to base rate
Accuracy	81.38%	79.84%	-1.5%	LR better (misleading metric)
True Positives	325	384	+59	More fraud caught
False Positives	36,973	40,098	+3,125	More false alarms
False Negatives	275	216	-59	Fewer missed frauds

Table 8. Full performance comparison: Logistic Regression vs. Random Forest (test set, n=200,000).

5.3 Correlation Analysis and Feature Design

All individual feature correlations with the fraud label are very weak ($|r| < 0.05$). The strongest is amount_log at $r = +0.044$, with velocity_score at $r = -0.002$ and amount_vs_mean_ratio at $r = -0.002$. This near-zero correlation structure is not a limitation — it is evidence of sophisticated evasion. Fraudsters have adapted to single-variable detection by matching legitimate users on individual dimensions (velocity, amount ratio) while deviating on multi-dimensional combinations (high amount + midnight + mobile). This is precisely the regime where ensemble machine learning models excel over rule-based systems, and it validates the AUC-ROC improvement observed when moving from LR to RF.

6. DISCUSSION

6.1 The Adversarial Adaptation Hypothesis

The collective findings of this paper support a coherent interpretation: Nigerian banking fraud is increasingly adversarially adaptive. The temporal velocity shift—slow during monitored hours, fast at midnight—demonstrates that fraudsters are aware of and actively circumvent bank monitoring cycles. The amount-camouflage pattern (fraudsters targeting amounts near users' historical averages) demonstrates awareness of amount-deviation detection rules. The near-zero correlations between individual features and fraud labels demonstrate successful adaptation to univariate statistical anomaly detection.

Banks that continue to rely on static rule-based systems are operating on the assumption that fraudsters are naive. The data does not support that assumption.

6.2 The Precision-Recall Tension

The 0.3% base rate creates an irreducible tension between fraud detection rate and false alarm rate that no model can fully resolve. At the Random Forest's operating point, 99.05% of flagged transactions are legitimate — a ratio that would overwhelm any manual review team and impose significant customer friction if acted upon directly. This suggests a tiered deployment strategy: ML models should drive risk scoring rather than binary block/pass decisions. High-risk scores trigger additional authentication (OTP, biometric step-up); medium scores trigger soft friction (SMS notification, transaction delay); only the highest-confidence fraud predictions warrant automatic blocking. This architecture preserves fraud detection benefits while managing customer experience costs.

6.3 Limitations

- Synthetic dataset: While distributional properties match known NIBSS statistics, the data was simulated. Findingsrequire validation on live production transaction streams before deployment decisions are made.
- Single time window: Seasonal fraud dynamics, year-over-year trend analysis, and the impact of specific CBN policyevents (naira redesign, cashless policy enforcement) cannot be assessed.
- Feature scope: Device fingerprinting, IP geolocation, SIM swap history, and biometric signals — all commerciallyavailable features in production fraud systems — were not included.
- No SHAP analysis: Feature interaction effects (e.g., high amount × midnight × mobile) were not decomposed. SHAPvalues would strengthen the interpretability of the Random Forest model for regulatory submissions.
- Composite risk mystery: The negative correlation between composite_risk and fraud label warrants furtherinvestigation — it may indicate feature construction issues or genuine fraudster adaptation to composite rules.

7. POLICY RECOMMENDATIONS

Based on the empirical findings, we offer the following prioritized recommendations for Nigerian commercial banks, structured by implementation horizon:

Immediate (0–3 months)

- Mandatory OTP for all Web and Mobile transactions above ₦50,000. Web fraud operates at 0.343% with no physical authentication barrier; OTP reduces card-not-present vulnerability directly.
- Dynamic nighttime velocity rules. Tighten transaction velocity thresholds between 11 PM and 4 AM, when fraudstervelocity spikes 45–58% above baseline.
- 24/7 customer support channels. Fraud victims currently wait until morning for resolution. Real-time reporting reduces fraudster window to complete follow-on transactions.
- Amount-based step-up authentication. Transactions above ₦500,000 on Mobile or Web channels should require a secondary verification factor, given 1.01% and 0.64% fraud rates at this tier.

Medium-term (3–6 months)

- Deploy ML risk scoring in production. Random Forest (AUC 0.82) should replace or augment existing rule-based systems, integrated as a risk score signal rather than a binary decision.
- Location-specific risk thresholds. Abuja (0.328% avg) and Rivers State (0.254%), warrant elevated sensitivity parameters relative to Lagos (0.234%).
- Mobile security hardening. Device binding, session anomaly detection, and in-app behavioural biometrics should be prioritized given mobile's automated fraud evidence.

Long-term (6–12 months)

- Investigate Oyo State POS concentration. FCMB × Oyo (1.087%) and multiple other bank-Oyo POS combinations show elevated rates suggesting a localised merchant network compromise.
- Industry fraud intelligence sharing. The near-uniform bank-level fraud rates (0.201%–0.284%) indicate fraudsters operate across institutions. NIBSS-coordinated real-time fraud signal sharing would benefit the entire ecosystem.
- 40+ user protection programmes. Despite lower per-transaction fraud rates, older users bear the largest absolute fraud burden. Simplified OTP flows and proactive SMS alerts are appropriate accommodations.

8. CONCLUSION

This paper has presented the first large-scale analysis of fraud patterns in Nigerian commercial banking using one million representative transactions. The findings collectively tell a story of sophisticated, adaptive fraud: fraudsters who time their attacks to coincide with sleeping victims, who slow their transaction velocity to evade speed-based detection, who target high-value transactions while mimicking legitimate amount profiles, and who exploit the mobile channel with apparent automation. None of these behaviours are detectable through simple univariate rules — they emerge only through multi-dimensional pattern analysis of the kind that machine learning enables.

The Random Forest classifier achieves AUC-ROC of 0.82 and detects 64% of fraud cases on Nigerian data, establishing a meaningful benchmark for future research. The precision-recall tension at 0.3% prevalence remains the central operational challenge, and we advocate for risk-score-based deployment architectures that apply tiered friction rather than binary block decisions.

For researchers, this work identifies a rich set of open problems: adversarial robustness testing against evasion-aware fraudsters, graph-based models to detect coordinated fraud rings across accounts, longitudinal analysis of fraud evolution under policy interventions, and cross-market comparison between Nigerian patterns and other emerging economies. For practitioners, the message is clear: Nigerian banking fraud is not random noise. It is strategic behavior that demands an equally strategic, data-driven response.

REFERENCES

[1] Abdallah, A., Maarof, M. A., & Zainal, A. (2016). Fraud detection system: A survey. *Journal of Network and Computer Applications*, 68, 90–113.

[2] Bahnsen, A. C., Stojanovic, A., Aouada, D., & Ottersten, B. (2016). Feature engineering strategies for credit card fraud detection.*Expert Systems with Applications*, 51, 134–142.

[3] Bhattacharyya, S., Jha, S., Tharakunnel, K., & Westland, J. C. (2011). Data mining for credit card fraud: A comparative study. *Decision Support Systems*, 50(3), 602–613.

[4] Blagus, R., & Lusa, L. (2017). SMOTE for high-dimensional class-imbalanced data. *BMC Bioinformatics*, 14(1), 106.

[5] Bolton, R. J., & Hand, D. J. (2002). Statistical fraud detection: A review. *Statistical Science*, 17(3), 235–249.

[6] Carcillo, F., Dal Pozzolo, A., Le Borgne, Y. A., Caelen, O., Mazzer, Y., & Bontempi, G. (2018). SCARFF: A scalable framework forstreaming credit card fraud detection with Spark. *Information Fusion*, 41, 182–194.

[7] Central Bank of Nigeria. (2023). *Annual Report on Payment System Fraud*. CBN Publications.

[8] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

[9] Dal Pozzolo, A., Caelen, O., Johnson, R. A., & Bontempi, G. (2015). Calibrating probability with undersampling for unbalancedclassification. In *2015 IEEE SSCI* (pp. 1–8).

[10] Elkan, C. (2001). The foundations of cost-sensitive learning. In *Proceedings of IJCAI 2001* (Vol. 17, pp. 973–978).

[11] GSMA Intelligence. (2023). *Mobile Money Fraud in Sub-Saharan Africa: Patterns and Prevention*. GSMA.

[12] Kou, Y., Lu, C. T., Sirwongwattana, S., & Huang, Y. P. (2004). Survey of fraud detection techniques. In *IEEE ICNSC 2004* (pp. 749–754).

[13] NIBSS. (2024). *Electronic Fraud Landscape Report Q4 2024*. Nigerian Inter-Bank Settlement System.

[14] Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifierson imbalanced datasets. *PLOS ONE*, 10(3), e0118432.

[15] Zhang, X., Han, Y., Xu, W., & Wang, Q. (2022). HOBA: A novel feature engineering methodology for credit card fraud detection with adeep learning architecture. *Information Sciences*, 557, 302–316.

[16] Owolabi, J. E. (2025). *Supervised Learning Model for Fraud Risk Scoring in Nigerian Financial Transactions*. BSc Statistics Dissertation, University of Lagos. Dataset available at: https://www.kaggle.com/datasets/hendurhance/nibsss-fraud-dataset (CC BY-SA 4.0).

[17] NIBSS. (2023). *Electronic Fraud Landscape Report: Full Year 2023*. Nigerian Inter-Bank Settlement System. https://nibss-plc.com.ng

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